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Fishing reel

Abstract

A level winding device for a fishing reel comprises a tubular body that rotates in conjunction with ON/OFF operation of a clutch mechanism, a worm shaft that is accommodated in the tubular body and rotationally driven by a rotational operation of a handle, a holding body that comprises an engagement pin engaged with a spiral groove of a worm shaft and holds a fishing line insertion portion through which a fishing line is inserted, and two circumferential detent portions (a rib and a recessed portion) provided in the holding body and the tubular body such that two bodies rotate integrally. The two circumferential detent portions are provided on the tubular body at an interval of approximately 180°, and an axial detent portion that suppresses rattling of the holding body in an axial direction of the tubular body is provided at an intermediate position between the two circumferential detent portions.

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Patent No.	Application Date	Country	CPC
2014-000045	12/2013	JP	N/A
WO-2013180061	12/2012	WO	A01K 89/01912

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2022-197745 filed on Dec. 12, 2022 in the Japanese Patent Office, the entire contents of each hereby incorporated by reference.

BACKGROUND OF THE INVENTION

1. Technical Field

(2) The present invention relates to a fishing reel comprising a fishing line guide body that winds and guides a fishing line around a spool between side plates in front of the spool.

2. Description of the Related Art

(3) In the related art, in a double-bearing type fishing reel, a level winding device is disposed to evenly wind a fishing line around a spool rotatably supported between left and right side plates. The level winding device includes a fishing line guide body that reciprocates left and right in front

of the spool in conjunction with a winding operation of a handle rotatably arranged on one side plate side, and the fishing line can be evenly wound around the spool by inserting the fishing line into the fishing line guide body.

(4) A configuration is known in which the above-described fishing line guide body includes an opening in which an opening region is formed wide so as to reduce resistance at the time of releasing the fishing line and a narrow portion that guides the fishing line at the time of winding the fishing line, and the fishing line guide body is turned in conjunction with a switching operation of a clutch (see, for example, JP 2014-000045 A). In such a configuration, since the opening is located in front of the spool when the clutch is turned OFF, the fishing line wound around the spool during a casting operation does not receive a large resistance from the inner surface of the opening of the fishing line guide body, and stable casting can be performed while suppressing a decrease in a flying distance of a tackle. In addition, when the clutch is returned to ON, the narrow guide portion is located in front of the spool, so that the fishing line can be evenly wound around the spool.

(5) The fishing line guide body described above is attached to a holding body that is movable in an axial direction (left-right direction) and integrally turns together with a tubular body with respect to the tubular body that turns in conjunction with the clutch operation. The holding body is provided with an engagement pin that engages with a spiral groove of a worm shaft disposed in the tubular body, and when the worm shaft is rotationally driven together with a rotational operation of the handle, the fishing line guide body is movable in the axial direction by an engagement relationship between the spiral groove and the engagement pin.

(6) Further, the tubular body and the holding body are configured to be integrally rotatable. In the configuration disclosed in JP 2014-000045 A, ribs extending in the axial direction are formed at intervals of approximately 180° on the outer surface of the tubular body, and the ribs are fitted into recessed portions formed in the holding body along the axial direction to integrally rotate. That is, when the tubular body is rotationally driven by the clutch operation, the holding body (the fishing line guide body) is configured to be integrally rotationally movable together with the tubular body by the fitting relationship described above.

SUMMARY OF THE INVENTION

(7) In the double-bearing reel described above, the tubular body and the holding body (the fishing line guide body) can be integrally rotated by the fitting relationship between the two ribs and the recessed portions. The rotation of the holding body is performed when the fishing line is not wound around the spool, that is, when the clutch mechanism is turned ON and OFF, and the fishing line guide body is stopped in the axial direction. The holding body slides in the axial direction with respect to the tubular body by winding a handle in a state where the clutch mechanism is turned on.

(8) The holding body of the related art has a problem that the holding body tends to rattle in the axial direction (a rattling that swings as indicated by arrows D2 and D3 in FIG. 6) when reciprocally driven in the axial direction (left-right direction). Then, when the holding body rattles in this manner, the fishing line cannot be neatly wound around the spool at the time of winding operation of the fishing line, and resistance is generated at the time of releasing the next fishing line. In addition, since the holding body rattles, an abnormal noise occurs at the time of winding operation, particularly at the time of winding operation at a high speed, and a sense of precision is lost.

(9) The present invention has been made in view of the above problems, and an object of the present invention is to provide a fishing reel having a configuration in which a fishing line guide body is swung, in which rattling of the fishing line guide body in an axial direction is suppressed during a winding operation of a fishing line.

(10) In order to achieve the above object, a fishing reel according to the present invention comprises: a spool rotatably supported between left and right side plates of a reel main body; a clutch mechanism that switches the spool between a fishing line winding state and a fishing line releasing state; and a level winding device that evenly winds a fishing line around the spool when

the clutch mechanism rotates a handle in an ON state, in which the level winding device comprises a tubular body rotatably supported between left and right side plates of a reel main body and configured to rotate in conjunction with ON and OFF operation of the clutch mechanism, a worm shaft accommodated in the tubular body and rotationally driven by a rotational operation of the handle, a holding body comprising an engagement pin engaged with a spiral groove of the worm shaft and configured to hold a fishing line insertion portion through which a fishing line is inserted, and two circumferential detent portions provided in the holding body and the tubular body so that the holding body and the tubular body integrally rotate, the two circumferential detents are provided on the tubular body at an interval of approximately 180°, and an axial detent portion that prevents the holding body from rattling in the axial direction of the tubular body is provided at an intermediate position between the two circumferential detents.

(11) According to the fishing reel having the above-described configuration, the tubular body constituting the level winding device and the holding body holding the fishing line insertion portion integrally rotate together with the clutch operation by the two circumferential detent portions. The two circumferential detent portions are provided on the tubular body at an interval of approximately 180°, and an axial detent portion that suppresses rattling of the holding body in the axial direction of the tubular body is provided at an intermediate position thereof. Therefore, when the fishing line is wound around the spool, the holding body is prevented from rattling along the axial direction even if the holding body holding the fishing line insertion portion is reciprocally driven between the left and right side plates. As described above, since the holding body does not rattle along the axial direction, the fishing line is accurately wound around the spool, and generation of the abnormal noise due to rattling is suppressed.

(12) According to the present invention, in a fishing reel having a configuration in which a fishing line guide body is swung, it is possible to obtain a fishing reel in which rattling of the fishing line guide body in an axial direction is suppressed during a winding operation of a fishing line.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view illustrating an embodiment of a fishing reel according to the present invention as viewed from a front side;
- (2) FIG. 2 is an exploded perspective view illustrating a structure of a fishing line guide body portion of the present embodiment;
- (3) FIG. 3 is a partial cross-sectional view illustrating a fishing line guide body;
- (4) FIG. 4 is a schematic view illustrating a configuration of a tubular body;
- (5) FIG. 5A is a front view for explaining the action of the fitting relationship illustrated in FIG. 4;
- (6) FIG. 5B is a side view of FIG. 5A;
- (7) FIG. 6A is a front view for explaining the action of the fitting relationship of the related art;
- (8) FIG. 6B is a side view of FIG. 6A; and
- (9) FIG. 7 is a schematic view illustrating another configuration of the tubular body.

DETAILED DESCRIPTION

(10) Hereinafter, an embodiment of a fishing reel according to the present invention will be described with reference to FIGS. 1 to 5B.

(11) Note that in the following description, a front-rear direction (front side, rear side), a left-right direction, an up-down direction (upper side, lower side) are defined as directions illustrated in FIG. 1.

(12) As illustrated in FIG. 1, the fishing reel according to the present embodiment comprises a reel main body 1 having left and right side plates 1A and 1B in which left and right frames 2a and 2b are covered with left and right covers 3a and 3b, respectively. In the reel main body 1, reel legs

located between the left and right side plates and attached to a fishing rod **100** are integrally formed. A spool shaft is rotatably supported between the left and right frames **2a** and **2b** via a bearing, and a spool **3**, around which a fishing line is wound, is integrally fixed to the spool shaft. (13) In the present embodiment, a handle **5** that rotationally drives the spool **3** is installed on the left side plate **1A** side, and a known power transmission mechanism that transmits a rotational driving force of the handle **5** to the spool shaft is disposed in a space between the left frame **2a** and the left cover **3a**. A known clutch mechanism that switches the spool shaft between the power transmission state and the power interruption state is disposed on one of the side plates, and the clutch mechanism switches from the clutch-ON state (power transmission state) to the clutch-OFF state (power interruption state) by pushing down a clutch lever **8** disposed between the left and right side plates on the rear side of the spool **3**. The return from the clutch-OFF state to the clutch-ON state can be performed by rotationally operating the handle **5** by a known automatic return mechanism.

(14) A level winding device **10** is disposed on the front side of the spool **3** between the left and right side plates **1A** and **1B**. The level winding device **10** comprises a fishing line guide body **20** through which the fishing line wound around the spool **3** is inserted, a worm shaft (screw shaft) **11** rotationally driven through the driving force transmission mechanism, and a tubular body (cylindrical body) **12** that accommodates the worm shaft **11**, and is configured to be reciprocally driven to the left and right when an engagement pin provided on the fishing line guide body **20** is engaged with the worm shaft **11**.

(15) The worm shaft **11** is accommodated in the tubular body **12** rotatably held between the left and right side plates, and a long hole **12a** extending in the axial direction is formed in an outer surface of the tubular body **12**. A spiral groove **11a** is formed on the surface of the worm shaft **11**, and the spiral groove **11a** is exposed through the long hole **12a**. In addition, the fishing line guide body **20** comprises a holding body (main body) **21** integrally formed of resin or the like, and a fishing line insertion portion **22** integrally fixed to an upper portion of the holding body **21** and through which the fishing line is actually inserted. A through hole **21a** is formed in the holding body **21** along the left-right direction, and is disposed and formed so as to surround the tubular body **12**. That is, the tubular body **12** is inserted through the through hole **21a** of the holding body **21**.

(16) The tubular body **12** is rotatably driven by a predetermined angle in accordance with the operation of the clutch lever **8** of the clutch mechanism. Specifically, when the clutch lever **8** is pushed down from the clutch-ON state illustrated in FIG. 1 to the clutch-OFF state, the tubular body **12** is rotatably driven to the front side (a direction of arrow **D1** in FIG. 1). Note that the power transmission mechanism that rotates and drives the tubular body **12** when the clutch lever **8** is pushed down can be performed by, for example, a power transmission mechanism similar to the configuration disclosed in JP 2014-000045 A described above. That is, when the clutch lever **8** is pushed down, the clutch plate (not shown) is rotationally driven, and the tubular body **12** is rotationally driven via the engagement protrusion (not illustrated) of the clutch plate engaged with the long hole **12B** of the rotation plate **12A** non-rotationally fixed to the tubular body **12**.

(17) The holding body **21** and the tubular body **12** are connected so as to be integrally rotatable by a circumferential detent portion. As illustrated in FIG. 3, the circumferential detent portion of the present embodiment is configured by forming two ribs **12b** protruding in a radial direction in the tubular body **12**, and forming two corresponding recessed portions **21b** such that the ribs **12b** are fitted to the holding body **21**. The rib **12b** and the recessed portion **21b** are formed at an interval of about 180° (including about 180°±20°), so that the holding body **21** is integrally rotationally driven together with the tubular body **12**. Due to the fitting relationship between the rib **12b** and the recessed portion **21b**, the holding body **21** is guided to reciprocate in the left-right direction with respect to the tubular body **12** during the fishing line winding operation.

(18) The holding body **21** comprises a cylindrical portion (configured in a cylindrical shape in the present embodiment) **21A** that protrudes toward the front side and configures a front end portion of

the level winding device **10**, and an engagement pin **23** that engages with a spiral groove **11a** formed in a surface of the worm shaft **11** is accommodated in the holding body through a long hole **12a** formed in the tubular body **12**.

(19) The engagement pin **23** comprises an engagement portion **23a** that engages with the spiral groove **11a** and a main body **23b** that is press-fitted into the cylindrical portion **21A**. The engagement pin **23** is configured to be retained in a state of being press-fitted into the cylindrical portion **21A**. As illustrated in FIG. 2, this retaining structure is configured by forming a through-hole **21d** in a diameter direction of the cylindrical portion **21A**, press-fitting the main body **23b** of the engagement pin **23** into the cylindrical portion **21A**, and press-fitting a retaining pin **25** in a state where the engagement portion **23a** is engaged with the spiral groove **11a**. In the present embodiment, the weight of the engagement pin **23** is reduced by forming a cavity portion **23c** in the main body **23b**, but the main body **23b** may be configured in a solid state.

(20) Furthermore, an O-ring **30** is attached to the cylindrical portion **21A** in correspondence with a press-fitting position of the retaining pin **25** so that the retaining pin **25** does not come out in the radial direction. A fixing groove **21B** is formed in the cylindrical portion **21A** along the circumferential direction, and the retaining pin **25** is prevented from being detached from the cylindrical portion **21A** by fitting the O-ring **30** into the fixing groove **21B**.

(21) The fishing line insertion portion **22** is fixed to the upper surface side of the holding body **21** by a fixing screw **28** or the like. Similarly to JP 2014-000045 A described above, the fishing line insertion portion **22** comprises an opening **22A** in which an opening region is formed wide so as to reduce resistance at the time of releasing the fishing line, and a narrow portion **22B** that guides the fishing line at the time of winding the fishing line, and is formed of a material having low fishing line resistance such as, for example, SUS or titanium. When the clutch is set to the OFF state, the holding body **21** is turned together with the tubular body **12**, and the opening **22A** is located in front of the spool **3** (fishing line releasing state), and when the clutch is set to the ON state, the holding body **21** is turned in the opposite direction together with the tubular body **12**, and the narrow portion **22B** is located in front of the spool **3** (fishing line guiding state).

(22) According to the level winding device **10** having the configuration described above, when the clutch mechanism is turned OFF at the time of the casting operation, the opening **22A** in which an opening region is formed wide is located in front of the spool, and thus the released fishing line does not receive a large resistance from the inner surface of the opening **22A**, and stable casting can be performed while suppressing a decrease in the flying distance of the tackle. When the clutch mechanism is returned to ON, the narrow portion **22B** is located in front of the spool (see FIG. 1). When the handle **5** is wound in this state, the fishing line guide body **20** (the fishing line insertion portion **22**) is reciprocally driven in the left-right direction by the engagement pin **23** engaged with the spiral groove **11a** of the rotatably driven worm shaft **11**, so that the fishing line is evenly wound around the spool **3**.

(23) The rib **40** provided on the fishing line guide body **20** is integrally provided at the center of the holding body **21** so as to protrude forward. By forming such a rib **40**, the fishing line is prevented from entering the gap or the stepped portion in the upper region of the cylindrical portion **21A** to cause line entanglement.

(24) The rib **40** is formed in a shape in which the fishing line that has caused the line trouble or the like easily comes off when coming into contact with the cylindrical portion **21A** and the vicinity thereof (a shape in which the fishing line slides in a direction in which the fishing line comes off). That is, as illustrated in FIGS. 2 and 3, the front end edge **40a** is inclined so as to approach the central portion of the holding body **21** as it transitions from the cylindrical portion **21A** side to the upper side (the fishing line insertion portion **22** side), whereby the fishing line can be easily pulled out upward.

(25) Furthermore, in the present embodiment, the O-ring **30** is attached to the cylindrical portion **21A** in correspondence with the press-fitting position of the retaining pin **25** so that the retaining

pin **25** does not come off in the radial direction. The fishing line may enter a gap between the O-ring **30** and the lower surface **40b** of the rib **40** to cause line entanglement. Therefore, the O-ring **30** is preferably disposed so as to fill a gap between the lower surface **40b** of the rib **40** of the cylindrical portion **21A** and an opposing portion **21e** facing each other in relation to the rib **40**.

(26) In the present embodiment, as described above, the fixing groove **21B** for press-fitting and fixing the O-ring **30** is formed in the cylindrical portion **21A** along the circumferential direction, so that the O-ring **30** is less likely to detach from the cylindrical portion **21A**. The fixing groove **21B** is not formed over the entire circumference of the cylindrical portion **21A**, but is preferably formed excluding a region (opposing portion **21c**) that becomes the opposing portion of the rib **40**. As described above, since the groove is not formed in one part thereof, the fixing operation of the O-ring **30** can be easily performed, and the fishing line will not be caught since the fixing groove does not exist at a portion where the fishing line easily enters.

(27) As described above, the holding body **21** and the tubular body **12** are formed with the circumferential detent portion (two ribs **12b** formed in tubular body **12** and two recessed portions **21b** formed in holding body **21**) at an interval of approximately 180° , whereby both members are configured to be integrally rotatable. That is, by forming two circumferential detent portions at an interval of approximately 180° on the outer peripheral surface of the tubular body **12** and the inner peripheral surface of the holding body **21**, when the tubular body **12** is rotated by operating the clutch mechanism, the holding body **21** (the fishing line insertion portion **22**) can be rotated integrally with the tubular body **12**.

(28) In the present invention, the axial detent portion **50** is provided at an intermediate position between the two circumferential detent portions so as to prevent the holding body **21** from rattling in the axial direction of the tubular body **12**. The axial detent portion **50** prevents rattling along the axial direction when the holding body **21** reciprocates in the horizontal direction via the worm shaft **11** and the engagement pin **23** through the winding operation of the handle. That is, in the configuration of the related art, as illustrated in FIGS. **6A** and **6B**, since only the circumferential detent portions are provided at intervals of approximately 180° , rattling (a rattling that swings around the holding body **21**) occurs along the axial direction. This rattling occurs mainly when the tension of the fishing line acts on side surfaces **22B1** and **22B2** of the narrow portion **22B** of the fishing line insertion portion **22** during the winding operation of the fishing line.

(29) Specifically, when the tension of the fishing line acts on the side surface **22B1** of the narrow portion **22B** along with the movement of the holding body **21** in the left-right direction, a force in the D2 direction acts on one end side of the holding body **21**, which causes rattling. In addition, when the tension of the fishing line acts on the side surface **22B2** of the narrow portion **22B**, a force in the D3 direction acts on the other end side of the holding body **21**, which causes rattling. In this case, if a gap G is generated between the inner peripheral surface of the holding body **21** and the outer peripheral surface of the tubular body **12** due to dimensional tolerance, the rattling becomes large.

(30) Therefore, by providing the axial detent portion **50** that prevents the holding body **21** from rattling along the axial direction at the intermediate position between the two circumferential detent portions so as to suppress the rattling described above, it is possible to suppress the holding body **21** from rattling in the axial direction of the tubular body **12**. In the present embodiment, as illustrated in FIGS. **4** to **6B**, the axial detent portion **50** is configured by a convex portion **12g** formed on the outer peripheral surface of the tubular body **12** and a concave groove **21g** formed on the inner peripheral surface of the holding body **21**.

(31) By providing such an axial detent portion **50**, the movement of the holding body **21** in the D2 and D3 directions is restricted by an upper surface **12g'** of the convex portion **12g** and a bottom surface **21g'** of the concave groove **21g** abutting on the rattling acting on the holding body **21** at the time of the fishing line winding operation as described above, and the rattling due to the movement in the D2 and D3 directions as illustrated in FIGS. **6A** and **6B** can be suppressed. That is, when the

fishing line is wound around the spool, rattling along the axial direction is suppressed even if the holding body **21** holding the fishing line insertion portion **22** is reciprocally driven between the left and right side plates. Therefore, the fishing line can be accurately wound around the spool, and generation of the abnormal noise due to the rattling is also suppressed, so that the fishing reel can be improved in a sense of precision. Furthermore, by providing the axial detent portion **50**, the turning operation of the fishing line guide body can be more stably performed.

(32) The above-described position where the axial detent portion **50** is provided may be an intermediate position between the two circumferential detent portions, but the axial detent portion is preferably provided within a range R separated by 45° or more in the circumferential direction from each circumferential detent portion (the position where the rib **12b** is formed) (see FIG. 4). By providing the axial detent portion **50** within such a range R, rattling can be stably suppressed. In particular, the axial detent portion **50** is preferably provided in a range facing the engagement pin **23** engaged with the spiral groove of the worm shaft **11**. That is, when the engagement pin **23** is engaged with the spiral groove **11a** of the worm shaft **11** through the long hole **12a** of the tubular body **12**, a large force is generated along the axial direction at this engagement portion. Therefore, as illustrated in FIG. 4, by providing the axial detent portion **50** (the convex portion **12g**) on the line Y connecting the center of the engagement pin and the axis C of the tubular body **12**, rattling can be more effectively suppressed.

(33) In fact, when a comparative test was conducted on the occurrence of rattling under the same conditions for the configuration in which the axial detent portion **50** was not provided (FIGS. 6A and 6B) and the configuration in which the axial detent portion **50** was provided (FIGS. 5A and 5B), it was confirmed that the rattling was reduced by 10% at the minimum and 35.9% at the maximum in the configuration in which the axial detent portion **50** was provided.

(34) FIG. 7 is a schematic view illustrating another configuration of the tubular body **12**. As shown in this embodiment, it is preferable that the convex portion **12g** constituting the axial detent portion **50** has a height H lower than each rib **12b** of the two circumferential detent portions, and a width W thereof is formed wider.

(35) As described above, by forming the convex portion **12g** of the axial detent portion **50** to have a small height and a large width, the concave groove can be made shallow to prevent a decrease in strength, and the contact area between the upper surface **12g'** of the convex portion **12g** and the bottom surface of the concave groove **21g** is increased, so that rattling can be more effectively suppressed.

(36) The embodiments according to the present invention have been described above, but the present invention is not limited to the embodiments and may be modified in various ways.

(37) The present invention has features in the portion of the fishing line guide body **20** (the holding body **21**) and the tubular body **12** of the level winding device **10**, and other configurations, for example, the size and shape of the reel main body, other functional members, and the like are not particularly limited. Furthermore, the configuration of the fishing line insertion portion **22**, the holding structure of the engagement pin engaged with the screw shaft, and the like can also be appropriately modified.

(38) In addition, two or more axial detent portions **50** may be provided as long as the axial detent portions are disposed at intermediate positions between the two circumferential detent portions. Further, when the axial detent portion **50** is formed in the fitting relationship between the convex portion **12g** and the concave groove **21g**, the axial detent portion can be appropriately deformed in terms of its height, width, groove depth, and the like.

Claims

1. A fishing reel, comprising: a spool rotatably supported between left and right side plates of a reel main body; a clutch mechanism that switches the spool between a fishing line winding state and a

fishing line releasing state; and a level winding device that evenly winds a fishing line around the spool when the clutch mechanism rotates a handle in an ON state, wherein the level winding device comprises a tubular body rotatably supported between left and right side plates of a reel main body and configured to rotate in conjunction with ON and OFF operation of the clutch mechanism, a worm shaft accommodated in the tubular body and rotationally driven by a rotational operation of the handle, a holding body comprising an engagement pin engaged with a spiral groove of the worm shaft and configured to hold a fishing line insertion portion through which a fishing line is inserted, and two circumferential detent portions provided in the holding body and the tubular body so that the holding body and the tubular body integrally rotate, the two circumferential detent portions are provided on the tubular body at an interval of approximately 180° , and an axial detent portion that prevents the holding body from rattling in the axial direction of the tubular body is provided at an intermediate position between the two circumferential detent portions.

2. The fishing reel according to claim 1, wherein the two circumferential detent portions comprise two ribs formed on an outer peripheral surface of the tubular body at an interval of approximately 180° and two recessed portions formed at corresponding positions of the holding body, and the axial detent portion is provided within a range separated by 45° or more in the circumferential direction from a position where the two ribs are formed.

3. The fishing reel according to claim 2, wherein the axial detent portion is provided within a range facing the engagement pin.

4. The fishing reel according to claim 2, wherein the axial detent portion is formed on an outer peripheral surface of the tubular body, and comprises a convex portion having a height lower than those of the two ribs and a width larger than those of the two ribs.
